Case Study Challenge Brief 2026: Sustainable Urban Mobility Infrastructure. Teams are required to evaluate the capital deployment feasibility and regulatory constraints of transitioning municipal transit networks to zero-emission battery electric bus fleets across Tier-1 metropolitan areas. Address total cost of ownership including depot charging infrastructure, grid interconnect upgrade timelines, lithium iron phosphate battery lifecycle degradation under extreme ambient thermal conditions, and route schedule optimization. Submissions must provide rigorous financial models and operational risk mitigation frameworks for municipal transport authorities.

Municipal Bus Depot Megawatt-Scale Fast Charging Interconnect Architecture:

Our analysis prioritizes 350-kilowatt direct current fast chargers configured in sequential overhead pantograph layouts at central bus garages. Utility tariff negotiations must establish dedicated industrial sub-stations to minimize coincident peak demand surcharges during evening replenishment windows.